

Overview

This assignment has two components:

1. **GitHub Classroom (Required):**
You will implement supervised learning models and generate required visualizations.
2. **Blackboard (Graded):**
You will upload selected plots and answer conceptual and interpretation questions based on your results.

Only the Blackboard component will be graded; however, completion of the GitHub Classroom component is *required* to verify that the work submitted to Blackboard was generated independently.

Important: Failure to complete the GitHub component may result in reduced or zero credit for Blackboard questions that rely on coding outputs.

Part I – GitHub Classroom

Goal

To independently implement and evaluate supervised learning models, including:

- ElasticNet regression
- Logistic regression
- k-Nearest Neighbors (kNN)
- Hyperparameter tuning
- Performance visualization using R^2 , AUROC, and AUPRC

The GitHub repository provides a structured environment and automated tests to verify successful implementation.

What You Will Implement

Regression Task

Using the Heart Disease dataset:

- Predict cholesterol (chol) using ElasticNet regression.
- Tune hyperparameters:
 - `alpha`
 - `l1_ratio`

- Evaluate performance using R^2 .
- Generate a heatmap of R^2 values across the hyperparameter grid.

Classification Task

Using the same dataset:

- Predict heart disease presence (num) as a binary classification problem.
- Train:
 - Logistic regression
 - kNN (tune `n_neighbors`)
- Evaluate using:
 - AUROC
 - AUPRC
- Generate:
 - AUROC curve for logistic regression
 - AUPRC curve for logistic regression
 - AUROC curve for kNN
 - AUPRC curve for kNN

Autograding & Verification

Automated tests run when you push your code. The autograder verifies:

- Models are implemented
- Required plots are generated
- Evaluation metrics are computed
- Outputs fall within valid numeric ranges

The GitHub component is not graded for code style, optimization, or efficiency. It is used to verify that:

- You independently implemented the required models
- The plots uploaded to Blackboard were generated from your own code

If the GitHub Classroom component is not completed:

- Marks associated with plot uploads and related short-answer questions may be deducted or withheld.
- Blackboard submissions containing plots without corresponding verified execution in GitHub may not receive credit.

AI Usage Disclosure (MANDATORY)

Your repository must contain a completed **AI_DISCLOSURE.md** file. You must disclose:

- Any AI tools used
- What tasks they assisted with
- How you verified correctness

Failure to complete this honestly and fully may result in a grade of zero.

Submission (GitHub Component)

Your GitHub submission is complete when:

- Required models are implemented
- All tests have run
- Required plots are generated
- AI disclosure is completed

You do not upload anything to Blackboard for Part 1.

PART 2 — Blackboard (Graded)

This portion evaluates your conceptual understanding and ability to interpret model performance. You must upload required plots and answer questions referencing your results. All written responses must be based on plots generated through your GitHub implementation.

Multiple Choice (3 points total)

Three multiple-choice questions assessing:

- ElasticNet regularization behavior
- Differences between AUROC and AUPRC
- Interpretation of metric discrepancies in imbalanced datasets

Each question is worth 1 point.

Short Answer/Plots (8 points for short answer questions + 10 points for plots)

Each response should be 2–4 sentences. Responses must reference your uploaded plots and include relevant metric values where requested/appropriate. General theory without reference to your results will not receive full credit.

GRADING SUMMARY

Blackboard (100%)

- Multiple Choice — 3 points
- ElasticNet Heatmap Upload — 2 points
- Classification Plot Uploads — 8 points
- Short Answer — 8 points

Total = 21 points

GitHub Classroom — Required (Verification Only)

- Verified model implementation
- Generated required outputs
- Completed AI Disclosure

Integrity & Expectations

This is an individual assignment.

- Code must be written independently.
- All outputs must be generated from your own implementation.
- GitHub autograding is used to verify execution.

You may discuss high-level concepts, but you may not share code or generated plots.

Tips for Success

- Tune hyperparameters systematically.
- Compare AUROC and AUPRC carefully as they measure different aspects of performance.
- Ensure plots are clearly labeled before submission.
- Keep written responses concise and specific.